

# MACHINE-CHECKED OBSTRUCTIONS FOR POTENTIAL-BASED APPROACHES TO THE COLLATZ CONJECTURE, WITH CLOSED-FORM BOUNDED BARRIERS

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**ABSTRACT.** We formalize, in Lean 4 over a single verified graph model — the admissible inverse cylinder tree of the accelerated Syracuse map — a complete obstruction map for the natural “local potential” proof strategies aimed at the divergence half of the Collatz problem, together with the strongest closed-form positive results those obstructions permit. The negative results are theorems, not heuristics: the per-block induction is provably circular; no congruence invariant separates start from goal at any finite modulus (every unit cylinder is reachable); no 3-adically continuous potential — in any codomain — certifies the barrier; and the linear bit-length class, which does certify the barrier in a closed form up to an exact bit criterion, dies at the very next step, even after being enriched with singular poles at the two attractors (an LP-duality argument whose rational Farkas certificates are themselves checked in Lean). The positive results include a closed-form bit-length potential proving the bounded  $\kappa$ -barrier without search, and a machine-checked formalization of the Eliahou–Simons–de Weger–Hercher cycle pipeline. The residual difficulty is isolated as a single named phenomenon — the archimedean Sturmian carry process of rotation number  $\log_2 3$  — for which the function-field analogue over  $\mathbb{F}_2[x]$ , here formalized end-to-end including a new sharper “plateau telescope” proof, serves as a control experiment. All headline theorems depend only on the three standard Lean axioms. The Collatz conjecture itself remains, of course, wide open; the contribution is a machine-checked map of where its difficulty does and does not live.

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The Lean formalization was developed in an extended human–AI collaboration with the assistant Claude (Anthropic); see §10 for the methodology and the division of labour, and the Acknowledgements. The artifact: <https://github.com/benetandujar72/collatz-admissible-tree>.

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## 1. INTRODUCTION

The Collatz (or  $3n + 1$ ) conjecture asserts that iterating  $n \mapsto n/2$  ( $n$  even),  $n \mapsto 3n + 1$  ( $n$  odd) from any positive integer eventually reaches 1. It famously splits into two independent halves: *no nontrivial cycles*, and *no divergent orbits*. Both are open; for the state of the art see Lagarias’s surveys [12, 13], Tao’s almost-everywhere result [21], the cycle exclusions of Simons–de Weger [18] and Hercher [8], and the computational verification frontier [1, 16].

This paper reports on a formalization programme (sessions S189–S243 of an ongoing human–AI collaboration) that attacks the problem from the *inverse* direction: instead of following orbits forward, it studies the tree of 3-adic cylinders that can reach a fixed target cylinder, and asks for *local potential functions* certifying that “cheap” inverse paths do not exist. The programme produced positive results — closed-form barriers — but its principal output is sharper and, we believe, more useful: a *complete, machine-checked obstruction map* showing that every natural potential-shaped strategy on this graph fails, for identified, theorem-level reasons.

Why publish negative results? Because rigorous obstruction knowledge redirects effort: a verified “this cannot work, and here is the certificate” spares a community years of unrecorded failed attempts, as recent machine-assisted refutations in extremal combinatorics have illustrated. The Collatz literature is dense with potential and invariant attempts; the map below makes a large class of them obsolete *as theorems*, each failure carrying a finite, independently re-checkable witness.

**Contributions.** All results are formalized in Lean 4 [14] over Mathlib [15]; the artifact builds with *zero sorry*. Headline theorems (Lean names in typewriter font; the axiom audit is Appendix A):

- (1) **The per-block induction is circular** (§3): the natural induction step is logically equivalent to the statement being proven — the formalization exposes a single residual crux (`BlockBoundaryExists`) — and the natural devices for discharging that crux are *refuted* by an explicit astronomical one-edge ladder built in closed form (`ladder_exists`; Theorem 3.3).
- (2) **No congruence invariant exists** (§4): every unit residue class modulo every power of 3 is reachable from the start (`invReachable_units`); separating invariants at any finite modulus are therefore impossible.
- (3) **No 3-adically continuous potential exists, in any codomain** (§5): discrete codomains collapse to finite moduli, and for real codomains the two *attractor identities*  $4(y - 1) = 3(c - 1)$ ,  $2(y + 1) = 3(c + 1)$  force any continuity modulus  $\omega$  to obey  $m \leq \omega(22 + Q)$  (`no_decaying_modulus_corrector`).
- (4) **A closed-form positive frontier, with an exact machine-checked wall** (§6): the potential  $\Phi_{\text{bit}} = \lfloor (\text{size}(c) - 1)/4 \rfloor + j - Q$  proves the bounded  $\kappa$ -barrier under the bit criterion  $4(m+Q) + 1 \leq \text{size}(\text{start})$  (`kappa_bounded_barrier_bitlen`, faithful instances exactly tight at 73 and 105 bits); and *no* member of the linear class survives one step further (`no_linear_size_potential_m1_Q8`). Enriching the class with singular poles at the two attractors buys exactly one more step and dies (`no_hybrid_potential_m1_Q9`); the Farkas certificates are rational, six rows long, and checked in Lean.
- (5) **The cycle pipeline, formalized** (§7): unconditional non-existence of Syracuse cycles of period  $\leq 4$ ; the bilateral window  $m \log_2 3 < A < 2m$ ; a one-inequality *master corridor* valid without any cycle hypothesis; and the full Eliahou–Simons–de Weger–Hercher reduction with kernel-checked continued-fraction instances (periods  $< 359$  from verification up to  $2^{20}$ ; periods  $< 16\,266$  from  $2^{29}$ ), connected to the literature’s verification statements by an explicit bridge.
- (6) **The function-field control experiment, formalized end-to-end** (§8): the Collatz analogue on  $\mathbb{F}_2[x]$  of Hicks–Mullen–Yucas–Zavislak [9] is proven in Lean (`collatz_F2X`) — to our knowledge its first formalization — via a new *plateau telescope* identity that is sharper than the original multiplicative-order argument. The same four local laws that are exact over  $\mathbb{F}_2[x]$  are lossy over  $\mathbb{Z}$ ; the difficulty of the integer problem is thereby exhibited theorem-against-theorem inside one artifact.

**What this paper does not claim.** Nothing here proves or claims progress on the Collatz conjecture itself. The barriers of §6 are finite statements about a specific cylinder tower; the obstructions say that a family of proof *shapes* cannot work on this model. We consider the honest delimitation of scope a

feature of the machine-checked format: every theorem below means exactly what its Lean statement says, no more.

## 2. THE VERIFIED MODEL

**2.1. The accelerated map and admissible classes.** For odd  $n \in \mathbb{N}_{>0}$  let  $\text{Syr}(n) = (3n + 1)/2^{\nu_2(3n+1)}$  be the accelerated Syracuse map. Working modulo 9, the 2-adic valuation  $\tau(\alpha) := \nu_2(3\alpha + 1)$  of the odd step is forced on the six *admissible* classes  $X_9 = \{1, 2, 4, 5, 7, 8\}$  by the table

$$\tau : \quad 1 \mapsto 2, \quad 2 \mapsto 1, \quad 4 \mapsto 2, \quad 5 \mapsto 3, \quad 7 \mapsto 4, \quad 8 \mapsto 1$$

(`tau`, `InX` in `AdmissibleBasic.lean`);  $\tau$  as used throughout is the table value of the class, with  $\tau \leq 4$  on  $X_9$  (`tau_le_four`; classes 0, 3, 6 never occur on odd orbits).

**2.2. The inverse cylinder graph.** Fix a base precision  $R$ . Vertices are pairs  $v = (j, c)$  with  $j \in \mathbb{N}$  (the number of *zero-steps* consumed) and  $c$  a residue class modulo  $3^{R+j}$  (`InvVertex`). There are two edge families (`InvEdgeOne`, `InvEdgeZero`):

$$\text{one-edge: } (j, c) \rightarrow (j, c') \quad \text{iff} \quad c' \in X_9, \quad c \equiv 2^{\tau(c')} c' \pmod{3^{R+j}},$$

$$\text{zero-edge: } (j, c) \rightarrow (j+1, c') \quad \text{iff} \quad c' \in X_9, \quad 3c + 1 \equiv 2^{\tau(c')} c' \pmod{3^{R+j+1}}.$$

One-edges invert a pure doubling block; zero-edges invert one odd step and *refine the modulus by one 3-adic digit*. The  $\kappa$ -weights (`InvKappaPreciseEdge`) are

$$\kappa(\text{one-edge}) = +1, \quad \kappa(\text{zero-edge}) = \begin{cases} -1 & \tau(c') = 1, \\ 0 & \tau(c') \geq 2, \end{cases}$$

so that along any path  $\sum \kappa = \#\text{one-edges} - \#\{\tau=1 \text{ zero-edges}\}$ , the combinatorial shadow of the *defect*  $D(u) = A(u) - 2q(u)$  of the corresponding forward word (`InvPathCounts.weight_lower_bound`).

**2.3. Words, the translation identity, and the S202 direction.** An *admissible word*  $u$  over  $\{0, 1\}$  encodes a composition of inverse Collatz steps: 1 multiplies by the forced power  $2^\tau$ , 0 applies  $n \mapsto (2^\tau n - 1)/3$ . The accumulated action is affine: a word with parameters  $(A, q, B)$  —  $A = \sum \tau$ ,  $q$  the number of zeros — maps  $n_0 \mapsto (2^A n_0 - B)/3^q$ , and words run from  $n_0 = 1$  satisfy the *translation identity*  $3^q n + B = 2^A$  (`evalWord`, `Defect.S202_translation_identity`).

The programme's organizing example is the 26-letter word  $W_{202}$  with  $(A, q) = (43, 22)$ ,  $B_{202} = 919\,447\,060\,349$  and final value  $n = 251$ . Its *defect* is  $D = A - 2q = -1 < 0$ : it refutes the naive monotonicity hope  $A \geq 2q$  and opens the programme's asymptotic slope question ( $\liminf A/q \geq 2$ ? — subcritical slope along admissible words would be the inverse-tree shadow of slow forward growth). The fixed point of its affine action,

$$x^* = \frac{B_{202}}{2^{43} - 3^{22}} \approx 0.105 \notin \mathbb{Z},$$

is a *virtual cycle*: no integer realizes it (there is no actual 43-double, 22-odd-step cycle), but 3-adically it is a unit, and its 3-adic expansion defines a coherent tower of cylinders

$$C_m = \{ n \equiv a^* \pmod{3^R} \}, \quad R = 22m + 2,$$

$$a^* \equiv B_{202} (2^{43} - 3^{22})^{-1} \pmod{3^R},$$

the integers that follow  $W_{202}$  backward integrally for  $m$  rounds (each round consumes the word's  $q = 22$  digits of 3-adic precision; the  $+2$  is the mod-9 base). Pleasingly, at  $m = 1$  the canonical residue is exactly  $a^* = 1 + 3^{22}$  (`aS202_decomp`), of bit-size 35.

The tower  $C_m$  is thus the canonical *cycle-resonant direction* of the inverse tree: the region where cheap (low-defect) inverse returns to 1 would concentrate if they existed. The corpus records the resulting dichotomy as the (open) *S202 alternative* (`S202Cylinders.lean`): either every  $C_m$  is accessible from 1 at defect cost  $o(m)$  — giving admissible families of subcritical slope  $\liminf A/q \leq 43/22 < 2$  — or accessibility fails infinitely often and a new 3-adic obstruction governs the admissible inverse cover. The barriers studied below are the quantitative, budget-bounded face of the second horn: they assert that within a zero-step budget  $Q$  there are *no* cheap returns along this tower. We emphasize the honest scope:  $C_m$  is a model family — the hardest natural direction first — and barriers on it are neither necessary nor sufficient for Collatz; their role is that every potential or induction strategy on the inverse tree must already succeed here, which is what makes the obstruction map of §§3–6 informative for strategy design at large.

**2.4. The start, the goal, and the barrier.** The graph-level start is  $\text{Start}(m) = (0, a^* \bmod 3^R)$  (`InvStart`, `aS202`); a vertex is a *goal* if its cylinder contains 1, i.e.  $c \equiv 1 \pmod{3^{R+j}}$  (`InvVertex.IsGoal`).

**Definition 2.1** (the bounded  $\kappa$ -barrier; `S202_kappa_precise_barrier_bounded`). For  $m, Q \in \mathbb{N}$ , every  $\kappa$ -weighted inverse path from  $\text{Start}(m)$  to a goal with at most  $Q$  zero-edges has total  $\kappa \geq m$ .

The semantic content on the forward side is the *slope barrier*: every admissible word  $u$  with  $q(u) \leq Q$  landing in the  $m$ -th S202 cylinder has defect  $D(u) \geq m$  (`S202_slope_barrier_from_kappa_precise_potential_bounded`, via the verified forward-inverse correspondence `S212_forward`). Informally: there are no cheap inverse returns along this resonant 3-adic direction. These are finite, decidable-in-principle statements; their interest is as the minimal non-trivial quantitative expansion property the inverse-tree programme needs, and as the target all potential strategies below aim at.

**Theorem 2.2** (the potential method; `S202_kappa_precise_barrier_bounded_from_potential`). Let  $\Phi$  be any function from vertices to  $\mathbb{Z}$  (or  $\mathbb{Q}$ ,  $\mathbb{R}$ ) such that (E)  $\Phi(v) \leq \kappa(e) + \Phi(v')$  for every edge  $e : v \rightarrow v'$  in the slice  $j \leq Q$ ; (G)  $\Phi(g) \leq 0$  for every goal  $g$  with  $g.j \leq Q$ ; and (S)  $\Phi(\text{Start}(m)) \geq m$ . Then the bounded  $\kappa$ -barrier holds at  $(m, Q)$ .

Everything in §§3–6 is about which  $\Phi$  can exist.

**2.5. The mod-9 rigidity.** Two structural facts used throughout: the source class of every one-edge lies in  $\{4, 7\} \bmod 9$  (`oneEdge_source_mod9_in_four_seven`; equivalently  $2^{\tau(\alpha)}\alpha \equiv \{4, 7\} \pmod{9}$  for all  $\alpha \in X_9$ , `two_pow_tau_mul_mod9`); and the *deep-jump trap*: a one-edge raises  $\nu_3(c-1)$  by  $\geq 2$  iff its target is  $\equiv 1 \pmod{9}$ , forcing the source class 4 (`oneEdge_deepjump_iff`, `oneEdge_deepjump_source_four`). The trap will reappear, unplanned, inside the LP certificates of §6 — the two obstruction families are one mechanism.

### 3. OBSTRUCTION I: THE INDUCTION IS CIRCULAR

The natural uniform-in- $m$  strategy is induction on the precision level: cut each level- $(m+1)$  path at a distinguished vertex whose prefix projects to an  $m$ -level path (apply the induction hypothesis:  $\kappa_1 \geq m$ ) and whose suffix contributes  $\kappa_2 \geq 1$ . The formalization makes the folklore difficulty exact.

**Theorem 3.1** (the reduction, and its single crux; `kappaPathSplit_of_block`, `uniform_kappa_barrier_of_block`, `uniform_slope_barrier_of_block`). *Let  $\text{BlockBoundaryExists}(m, Q)$  be the statement: every level- $(m+1)$   $\kappa$ -path from the start to a goal of depth  $\leq Q$  admits a split point mid whose prefix projects onto an  $m$ -goal and whose suffix has  $\kappa_2 \geq 1$ . Then a base barrier at level  $m_0$  together with  $\text{BlockBoundaryExists}$  at every level  $\geq m_0$  yields the bounded  $\kappa$ -barrier — and hence the slope barrier  $D \geq m$  — at every level  $m \geq m_0$ .*

The entire open content of the route is thus the crux. Both natural ways of discharging it are dead:

**Theorem 3.2** (the unconditioned increment is false; `outerBlockIncrement_false`). *The statement “every suffix from a projected  $m$ -goal contributes  $\kappa_2 \geq 1$ ” is refutable outright (take the root vertex and the empty suffix).*

**Theorem 3.3** (the canonical devices are false). *The repaired devices — choosing the first  $m$ -precise vertex as the split, or requiring the split point itself to be  $m$ -precise — are also refutable, for all  $m, Q \geq 1$  (`not_firstMPrecisionSuffixPositive`, `not_blockBoundaryExists`).*

The refutation is constructive and, we think, the most instructive object of the negative half: a path consisting of one  $\tau \geq 2$  zero-edge (weight 0) followed by an astronomical *one-edge ladder* alternating between the classes 4 and 7 mod 9 (`ladder_exists`; the “mod-6 shield” `mod6_of_two_pow_mod9_seven` keeps every rung non- $m$ -precise), ending with a single deep one-edge into the goal (1, 1) (`deep_edge`). The ladder’s length is governed by  $\text{ord}(2 \bmod 3^{R+1}) = 2 \cdot 3^R$  (`orderOf_two_zmod_three_pow`) — about  $10^{22}$  rungs already at the  $m = 2$  precision — yet it exists *in closed form*: no enumeration appears anywhere in the proof. Its suffix after the forced split has

$\kappa_2 = 0 < 1$ , contradicting the devices. Notably, the same order-of-2 computation also disposes of all closed one-edge walks (`closed_oneWalk_weight_ge_order`): the graph has no short  $\kappa$ -cheap cycles to exploit.

*Remark 3.4.* The barrier itself survives Theorem 3.3 (the split is existential, and on the ladder path the prefix already pays  $\kappa_1 \geq m$ ); what dies is every *canonical* (choice-free) construction of the split. The induction route needs a genuinely new idea for selecting split points, or must be abandoned.

#### 4. OBSTRUCTION II: NO FINITE-MODULUS INVARIANT

A second classical hope is a congruence invariant: a union of residue classes modulo  $3^t$  (or any modulus) containing the start, closed under edges, and avoiding the goal.

**Theorem 4.1** (`reach = units`; `invReachable_units`). *For every  $m \geq 1$ , every  $t \geq 1$  and every  $z$  coprime to 3, some vertex reachable from `Start(m)` satisfies  $v.c \equiv z \pmod{3^t}$ .*

The proof is an induction on precision: each zero-edge contributes exactly one new 3-adic digit, and the lifting lemma `invReachable_zero_lift` (driven by `two_pow_tau_mul_mod9`) realizes any prescribed next digit. Consequently *no* congruence invariant separates start from goal at any finite modulus; in particular every discrete-codomain potential that factors through a finite modulus is dead. Even the “trap” cylinder of the deep-jump analysis is reached at every precision (`invReachable_hits_trap`).

#### 5. OBSTRUCTION III: NO CONTINUOUS POTENTIAL, IN ANY CODOMAIN

Could  $\Phi$  be 3-adically *continuous* — depending, with decaying influence, on deeper and deeper digits of  $c$ ? For  $\mathbb{Z}$ -valued  $\Phi$ , continuity means local constancy, hence factoring through some  $c \pmod{3^t}$ , dead by Theorem 4.1. The real-valued case is killed by two exact identities.

**Proposition 5.1** (the attractor identities; `tau2_attractor`, `tau1_attractor`, `tau2_attractor_nu3`, `tau1_attractor_nu3`). *Along a  $\tau = 2$  zero-edge with target  $y$ ,  $4(y - 1) = 3(c - 1)$ , so  $\nu_3(y - 1) = \nu_3(c - 1) + 1$ ; along a  $\tau = 1$  zero-edge,  $2(y + 1) = 3(c + 1)$ , so  $\nu_3(y + 1) = \nu_3(c + 1) + 1$ .*

Thus the  $\tau=2$ , class-1 chain converges 3-adically to the goal 1 while every edge on it has  $\kappa = 0$ , and the  $\tau=1$  filament converges to  $-1$  paying  $\kappa = -1$  per step. Moreover the start sits *on* the  $+1$  attractor shell:  $\nu_3(a^* - 1) = 22$  (`nu3_aS202_sub_one`; indeed  $a^* - 1 = 3^{22}$  exactly at  $m = 1$ ). Following the chain from the start for  $Q$  steps (`attractor_chain`) yields:

**Theorem 5.2** (no decaying-modulus corrector; `no_decaying_modulus_corrector`). *Let  $m \geq 1$  and let  $\Phi : \text{vertices} \rightarrow \mathbb{R}$  satisfy (E), (G), (S) of Theorem 2.2 together with a 3-adic continuity modulus  $\omega$ : equal  $j$  and  $c \equiv c' \pmod{3^t}$  imply  $\Phi(v) - \Phi(v') \leq \omega(t)$ . Then  $m \leq \omega(22 + Q)$ .*

Since this holds uniformly in  $Q$ , no fixed decaying modulus serves any  $m \geq 1$ : 3-adic continuity is incompatible with the contract, at every codomain. Numerical companions (Appendix C) quantify the failure: on certified finite regions, the empirical Hölder constant required at exponent  $\alpha$  scales as  $3^{\alpha(R+Q+\text{const})}$  — astronomically slack — and a Green-type singular part centred on the natural pole set does not repair it (the “interior” depth profile is robust to enlarging the pole family to  $2^s\{\pm 1, \pm 4, \pm a^*\}$ ,  $s \leq 8$ ). The tropical (min-plus) variant of the same idea is refuted by an explicit six-edge quotient-graph witness, and the optimal-path language is the period-one  $\tau=1$  filament rather than a Sturmian word — the rotation number  $\log_2 3$  enters globally, not locally (tool-level verdicts, tier T2/T3 of §10).

## 6. THE POSITIVE FRONTIER, AND ITS EXACT WALL

**6.1. The closed-form barrier.** The two integer *size laws* — with  $\text{size}(n)$  the bit length —

$$\text{size}(2^\tau n) = \text{size}(n) + \tau, \quad \text{size}(3n + 1) \geq \text{size}(n) + 1$$

(`size_two_pow_mul`, `size_three_mul_add_one`; the second is an inequality, and *all later difficulty lives in its slack*) give edge estimates for the canonical residues (`oneEdge_size_le`, `zeroEdge_size_le`) and hence:

**Theorem 6.1** (the bit-length potential; `PhiBitlen`, `kappa_bounded_barrier_bitlen`, `kappa_bounded_barrier_bitlen_from`). Define  $\Phi_{\text{bit}}(v) = \lfloor (\text{size}(v.c \bmod 3^{R+j}) - 1)/4 \rfloor + j - Q$ . Then  $\Phi_{\text{bit}}$  satisfies (E) and (G) unconditionally, and (S) holds whenever

$$4(m + Q) + 1 \leq \text{size}(\text{start residue}) + 4j_{\text{start}}.$$

Hence the bounded  $\kappa$ -barrier holds, in closed form and with no search, under the bit criterion.

Instances: at  $m = 1$  (35 bits) the criterion gives  $Q \leq 7$  (`kappa_barrier_m1_Q7`), subsuming the per-instance BFS certificates; a uniform corollary covers all  $m + Q \leq 8$  in one stroke (`kappa_barrier_of_sum_le_eight`); and for the *faithful* per- $m$  starts  $a_m^*$  (the Caveat-C-corrected tower, `as202_at`) the criterion is *exactly* tight:  $m = 2$ ,  $Q \leq 16$  with  $4(2+16)+1 = 73 = \text{size}(a_2^*)$ , and  $m = 3$ ,  $Q \leq 23$  with  $4(3+23)+1 = 105 = \text{size}(a_3^*)$  (`kappa_barrier_at_m2_Q16`, `kappa_barrier_at_m3_Q23`). All of these are kernel-checked on stored numerals; no compiled evaluation (`native_decide`) remains in this chain.

**6.2. The wall, machine-checked on both sides.** Is the bit criterion an artifact of  $\Phi_{\text{bit}}$ , or the boundary of its whole class? We answered this with a linear program over the feature coefficients, solved by constraint generation (the slices at the wall scales are far beyond practical enumeration, but the LP has five unknowns; the separation oracle walks the graph). Every infeasibility verdict carries an exact rational Farkas certificate — finitely many



true rows whose nonnegative combination is contradictory — re-verified independently against the raw modular arithmetic. The two  $m = 1$  certificates were then formalized:

**Theorem 6.2** (the linear class dies at  $Q = 8$ ; `no_linear_size_potential_m1_Q8`). *No  $\theta_0, \theta_1, \theta_2 \in \mathbb{Q}$  make  $\Phi(v) = \theta_0 + \theta_1 j + \theta_2 \text{size}(v.c \bmod 3^{24+j})$  satisfy (E), (G), (S) at  $(m, Q) = (1, 8)$ . The certificate has four rows — start, goal, one  $\tau=1$  zero-edge, one  $\tau=4$  one-edge — with multipliers  $(2, 2, 16, 17)/37$ .*

With Theorem 6.1 at  $Q = 7$ , the wall of the bit-length class sits *exactly* at the bit criterion, certified on both sides.

**Theorem 6.3** (attractor poles buy exactly one step; `no_hybrid_potential_m1_Q9`). *Enrich the class with the two singular features  $\nu_3(c-1)$  and  $\nu_3(c+1)$  (poles at the attractors of §5, the zero residue valued at full depth). The enriched LP is feasible at  $(1, 8)$  — the poles genuinely help — and infeasible at  $(1, 9)$ : no  $\theta \in \mathbb{Q}^5$  satisfies the contract. The six-row certificate has multipliers  $(280, 280, 2268, 2115, 404, 252)/5599$ , and its binding row is a  $\tau=2$  deep-jump one-edge (the mod-9 trap of §2:  $\nu_3(c-1)$  jumps  $1 \rightarrow 8$  at cost  $\kappa = +1$ ), which pins the pole coefficient.*

The numerical sweep extends the picture (tier T2; exact certificates at every dead point):

| class                        | $m = 1$ (35-bit start)             | $m = 2$ (73-bit faithful start)      |
|------------------------------|------------------------------------|--------------------------------------|
| linear $(1, j, \text{size})$ | alive $Q \leq 7$ / dead $Q \geq 8$ | alive $Q \leq 16$ / dead $Q \geq 17$ |
| + poles $\nu_3(c \mp 1)$     | alive $Q \leq 8$ / dead $Q \geq 9$ | alive $Q \leq 18$ / dead $Q \geq 20$ |

The mechanism is uniform: the poles' budget grows like  $(Q + 2) |\theta_{\text{pole}}|$ , but deep-jump trap edges cap  $|\theta_{\text{pole}}|$ , so the enrichment is asymptotically worthless. Honest scope: these theorems kill *feature-linear* classes against the potential contract of Theorem 2.2; a non-linear integer potential always exists when the barrier is true (the exact value function — e.g. the certified  $(1, 8)$  barrier itself, `Cert_m1_Q8_S216_Barrier`). The content is that no *closed form* of these shapes reaches it.

## 7. THE CYCLE HALF: THE ELIAHOU–HERCHER PIPELINE, FORMALIZED

The cycle half of the programme formalizes the classical reduction [6, 18, 8] in elementary, kernel-checkable form. Throughout, a (Syracuse)  $m$ -cycle is  $\text{Syr}^{[m]}(a_0) = a_0$  on odd  $a_0 > 1$  with period  $m \geq 1$ .

**Theorem 7.1** (small periods, unconditionally; `syr_no_nontrivial_fixedpoint` through `syr_no_nontrivial_4cycle`). *There is no nontrivial Syracuse cycle of period  $m \leq 4$ .*

**Theorem 7.2** (the bilateral window; `syr_cycle_window`). *Every nontrivial  $m$ -cycle satisfies  $m \log_2 3 < A < 2m$ , where  $A = \sum_{i < m} \nu_2(3a_i + 1)$ .*

The upper half comes from the multiplicative telescope  $2^A \prod a_i = \prod (3a_i + 1)$  (`syr_cycle_mul_equation`) and  $3a + 1 \leq 4a$ ; it needs no real analysis. A by-product worth recording is a *master corridor* requiring no cycle hypothesis at all (`syr_master_corridor`):

$$3^q a_0 < 2^{A_q} \text{Syr}^{[q]}(a_0) \leq 4^q a_0 \quad (a_0 \geq 1, q \geq 1),$$

one inequality whose two shadows are the cycle window and the trivial growth baseline.

The reduction machinery then follows the literature’s shape: the gap identity for  $|2^A - 3^m|$  (`syr_cycle_gap_eq`), the coupling of cycle elements to a *verification frontier* (`SyrVerifiedUpTo` as an explicit hypothesis; `syr_cycle_elements_above_frontier`, `syr_cycle_gap_mul_frontier_lt`), and the continued-fraction *mediant exclusion*: for a unimodular convergent pair  $L/\Lambda < \log_2 3 < U/\Upsilon$ , the pure- $\mathbb{N}$  identity  $m = \Lambda(Um - A\Upsilon) + \Upsilon(A\Lambda - Lm)$  (`mediant_period_bound`) turns the window into a period bound (`syr_mediant_exclusion`).

**Theorem 7.3** (kernel-checked instances; `syr_no_cycle_below_359`, `syr_no_cycle_below_16266`, `collatz_no_cycle_below_359`, `collatz_no_cycle_below_16266`). *Conditional on orbit verification up to  $2^{20}$  (resp.  $2^{29}$ ), there is no nontrivial Syracuse cycle of period below 359 (resp. 16266), via the convergent pairs  $84/53 < \log_2 3 < 485/306$  and  $1054/665 < \log_2 3 < 24727/15601$ . The same conclusions hold from verification of the standard (full) Collatz map, through the bridge `syrVerified_of_collatzVerified`.*

The 16266 instance involves  $\sim 150\,000$ -digit integers; a fuel-based binary-powering function (`fastPowAux`, `fastPow_eq`) makes the comparisons kernel-reducible at logarithmic recursion depth, so the whole pipeline is checked by the proof kernel with no compiled oracle. Convergent pairs are now mechanical; the ceiling of the method — closing *all* periods — is the open archimedean question (§8), for which the artifact provides a named socket (`EffectiveLinearForm`): any future effective linear-forms-in-logarithms bound plugs in as a hypothesis, not an axiom. (For comparison, the current literature record excludes  $m$ -cycles for  $m \leq 91$  [8], atop the computational verification frontier of [1]; our instances demonstrate the verified pipeline at deliberately small frontiers.)

## 8. THE DIFFICULTY, NAMED — AND THE FUNCTION-FIELD CONTROL

**8.1. One object behind four walls.** The residual obstructions of §§3–7 are not four problems but one. The slack of  $\text{size}(3n + 1) \geq \text{size}(n) + 1$  — equivalently the carry pattern of base-2/base-3 interaction — is the Sturmian process of rotation number  $\log_2 3$ : the sequence  $\text{size}(3^k) - \lfloor k \log_2 3 \rfloor$  and its orbit-level analogues. The wall of Theorem 6.2 sits where the linear bit budget meets this slack; the cycle ceiling of §7 is the effective rational approximation of  $\log_2 3$ ; the continuity no-go of Theorem 5.2 reflects that

the carry process is invisible 3-adically. Effective results on this object are scarce; the only unconditional anchor we know is Stewart’s theorem on digits in two bases [19] (see also [17]), and the Collatz process is known to embed base conversion [20].

**Question 8.1** (carry Mason–Stothers, weak form). Call an admissible-word step a *carry step* if  $\text{size}(3c + 1) = \text{size}(c) + 2$  (the excess case of the lossy size law). Is there an effective  $\delta > 0$  and  $\ell_0$  such that every admissible word of length  $\ell \geq \ell_0$  contains at least  $\delta \ell$  carry steps? Equivalently: an effective positive lower density of carries in the base-2 Sturmian coding of multiplication by 3 along admissible orbits. Any such bound strengthens the size laws coefficient-wise, pushing the closed-form barrier past the bit-criterion wall ( $Q \approx 7.7m$  on the faithful tower, §6) and breaking Theorem 6.2 from outside the linear class.

**8.2. The  $\mathbb{F}_2[x]$  twin, formalized end-to-end.** The control experiment makes the diagnosis precise. On  $\mathbb{F}_2[x]$  the Collatz analogue of Hicks–Mullen–Yucas–Zavislak [9] (see [3, 4, 2] for recent function-field developments, including general coefficient rings, completions, and sharp stopping times)

$$T(f) = f/x \quad (f(0) = 0), \quad T(f) = (x + 1)f + 1 \quad (f(0) = 1)$$

is a theorem: every nonzero  $f$  reaches 1. We formalized it completely (`F2Collatz.lean`; to our knowledge the first formalization), with the exact local laws displayed against their lossy integer twins:

| $\mathbb{F}_2[x]$ (exact; Lean theorem)      | $\mathbb{Z}$ (lossy; all difficulty here)       |
|----------------------------------------------|-------------------------------------------------|
| $\deg((x+1)f+1) = \deg f + 1$                | $\text{size}(3n+1) \geq \text{size}(n) + 1$     |
| $\deg(f/x) = \deg f - 1$                     | $\text{size}(n/2^\tau) = \text{size}(n) - \tau$ |
| odd step lands even                          | $3n + 1$ even                                   |
| $\deg(T^{[k]}f) \leq \deg f + 1 \ \forall k$ | <i>no integer counterpart</i>                   |

(Lean names, row by row: `oddStep_natDegree`, `evenStep_natDegree`, `oddStep_coeff_zero`, `collatzT_iterate_natDegree_le`.)

**Theorem 8.2** (the HMYZ theorem; `collatz_F2X`). *For every  $f \in \mathbb{F}_2[x]$ ,  $f \neq 0$ , there is  $k$  with  $T^{[k]}f = 1$ .*

Our proof improves on the original plateau argument. The two-step plateau map on odd  $f$  satisfies the *plateau telescope* (`plateau_step_identity`)

$$x(T^2f + 1) = (x + 1)(f + 1) :$$

in the  $f + 1$  coordinate the plateau is multiplication by  $(x+1)/x$ , so each plateau step consumes exactly one trailing factor  $x$  of  $f + 1$ , and the plateau exits within  $\text{ord}_x(f_0 + 1) - 1 \leq \deg f_0 - 1$  steps (`plateau_exit`) — pure ring algebra, no multiplicative-order or Frobenius argument needed.

The moral is sharpened by a falsity: the naive function-field analogue of Baker’s theorem fails over  $\mathbb{F}_2[x]$  (the Frobenius unit gaps  $x^{2^k} + (x + 1)^{2^k} = 1$  give infinitely many “small” gaps), and yet Theorem 8.2 stands. The

integer difficulty is therefore *not* reducible to “effective Baker is missing”: what the function field has and  $\mathbb{Z}$  lacks is carry-freeness of the size function. Question 8.1 is the precise residue of the comparison.

## 9. RELATED WORK

Surveys and general background: [12, 13, 23]. Forward-density and stochastic results: [22, 7, 10, 21]; our `SyracuseMeasure.lean` formalizes the support analysis of the Syracuse 2-adic law used in [21]. Cycles: [6, 18, 8]; verification: [16, 1]. Undecidability of generalizations: [5, 11]. Function fields: [9, 3, 4, 2]; base conversion: [20]; two-base digit results: [17, 19]. Formalization: Lean 4 [14] and Mathlib [15]; we are not aware of prior machine-checked obstruction maps for Collatz strategies, nor of prior formalizations of [9] or of the Eliahou-style cycle reduction.

## 10. METHODOLOGY, EVIDENCE TIERS, AND AXIOM HYGIENE

The collaboration ran as a funnel with three evidence tiers, and we report every claim with its tier:

- T1** *Lean theorems* over Mathlib, zero `sorry`, axiom set exactly `{propext, Classical.choice, Quot.sound}` (verified per headline theorem with `#print axioms`; Appendix A). All theorems cited in §§2–8 are T1.
- T1'** Lean theorems whose proofs invoke the compiled evaluator (`native_decide`), used *only* in the legacy per-instance BFS certificate family (`Cert_m1_Q8_S216_Barrier` and its Q10 sibling); the trusted base there includes the Lean compiler. No theorem in this paper’s main line depends on them; nine former `native_decide` anchors were re-proved kernel-only during the programme.
- T2** Exact-rational tool verdicts: LP Farkas certificates re-derived in exact arithmetic and re-verified against the model by an independent code path (the GATE-H sweep of §6; two representative certificates promoted to T1).
- T3** Numerical gates (profiles, frontiers, shape diagnostics) guiding the search; never load-bearing for a stated theorem.

The working method — a numerical gate first, adversarial verification second (two plausible intermediate claims produced by exploratory runs were *refuted* at this stage and never entered the corpus), Lean authentication last — kept the formal artifact small relative to the search space explored. The division of labour: the human author set the programme, the targets and the acceptance criteria; the AI assistant proposed and executed proofs, gates and refactorors; everything that survived is machine-checked, so trust reduces to the Lean kernel plus the stated axioms, not to either author.

Reproducibility: the repository contains the full Lean source (64 modules), the gate scripts and their JSON outputs (`tools/`), and build instructions

(narrow `lake build` targets; only the legacy `T1'` certificates are slow). A tagged commit accompanies the arXiv version.

## 11. OPEN PROBLEMS

- (1) **CMS** (Question 8.1), weak and orbit forms — the named residue of the whole map. Any effective carry density extends the closed-form barrier past the wall.
- (2) **The split crux**: a non-canonical construction of the predicate `BlockBoundaryExists` that survives Theorem 3.3, or a proof that no choice function over paths can supply it.
- (3) **Uniformity of the LP wall**: the GATE-H mechanism is identical at  $m = 1, 2$ ; a uniform-in- $m$  Lean theorem (“for every  $m$ , the hybrid class dies at  $Q = O(m)$ ”) looks within reach by hand.
- (4) **The cycle ceiling**: any effective linear-form bound plugs into `EffectiveLinearForm`; short of that, pushing convergent pairs is mechanical but unbounded.
- (5) **Beyond feature-linearity**: the obstruction map is design information — it says any viable potential must be non-linear in  $(\text{size}, j)$ , 3-adically discontinuous, and immune to the deep-jump trap. Characterizing the minimal such class is open.
- (6) **Function-field bonus**: the Frobenius–Catalan classification  $x^A + (x + 1)^m = 1 \iff A = m = 2^k$  over  $\mathbb{F}_2[x]$ , a clean candidate for Mathlib.

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## APPENDIX A. AXIOM AUDIT

Every theorem named in this paper was audited with `#print axioms` at its commit; each reports exactly `{propext, Classical.choice, Quot.sound}` — the three standard Lean axioms — with the single exception class noted as `T1'` in §10 (legacy BFS certificates, not used here). Selected audit rows:

| theorem                                           | module              |
|---------------------------------------------------|---------------------|
| uniform_slope_barrier_of_block                    | KappaSplitReduction |
| not_firstMPrecisionSuffixPositive                 | LadderExists        |
| not_blockBoundaryExists                           | LadderExists        |
| invReachable_units                                | ReachUnits          |
| no_decaying_modulus_corrector                     | AttractorNoGo       |
| kappa_bounded_barrier_bitlen, kappa_barrier_m1_Q7 | BitlenPotential     |
| kappa_barrier_at_m2_Q16, kappa_barrier_at_m3_Q23  | FaithfulBarrier     |
| no_linear_size_potential_m1_Q8                    | HybridPotentialNoGo |
| no_hybrid_potential_m1_Q9                         | HybridPotentialNoGo |
| syr_cycle_window, syr_master_corridor             | CycleReduction      |
| syr_no_cycle_below_16266                          | CycleReduction      |
| collatz_no_cycle_below_359 (and 16266)            | CollatzBridge       |
| collatz_F2X, plateau_exit                         | F2Collatz           |

## APPENDIX B. THE LEAN MODULE MAP

The artifact comprises 64 modules. Main groups (full map in the repository README):

- graph model and word correspondence: `AdmissibleBasic`, `InverseGraph`, `S202Cylinders`, `PathWordCorrespondence`, `S212Forward`, `AnalyticBarrier`;
- the obstruction map: `KappaSplitReduction`, `LadderExists`, `LadderRefutation`, `ReachUnits`, `OneEdgeMod9`, `AttractorNoGo`, `HybridPotentialNoGo`;
- the positive frontier: `BitlenPotential`, `FaithfulBarrier`, `AS202Lift`;
- arithmetic infrastructure: `DiscreteLog`, `LTEBound`, `RhoTransitionLaws`, `BridgeTruth`, `ClosedWalk`;
- the cycle half: `CycleEquation`, `CycleReduction`, `CollatzBridge`;
- the function-field twin: `F2Collatz`; the measure-theoretic support analysis: `SyracuseMeasure`; the legacy certificates: `Cert_*` (tier T1').

Build: `lake build CollatzLean4.<Module>` per module; the T1' certificates are the only slow targets.

## APPENDIX C. NUMERICAL GATES: SCRIPTS AND DATA

The `tools/` directory (committed, with JSON outputs) contains the gate scripts behind the T2/T3 claims, including: `phi_realkam_lp.py` (the Hölder-corrector LP with ultrametric dendrogram hops and exact witness re-verification), `phi_hybrid_gate.py` (the GATE-H constraint-generation LP; exact rational Farkas certificates; the source of Theorems 6.2–6.3), `phi_hybrid_interior.py` (pole-set robustness), `phi_tropical_*.py` (the tropical refutation and its witness), `sturmian_gate.py`, and the  $\sim 75$  archived research probes whose verdicts shaped the programme. Each script documents the epistemic status of each possible outcome.

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